

Lab 1 — PECARN TBI Data

Stat 214, Spring 2026 | February 17, 2026

1. Introduction

Traumatic brain injury (TBI) is a leading cause of pediatric morbidity in the US, with over 500,000 children presenting to emergency departments annually. The central clinical challenge is identifying which patients require CT imaging — unnecessary scans expose developing brains to radiation, while missed injuries can be fatal. The PECARN study (Kuppermann et al., 2009) developed age-stratified clinical decision rules to safely reduce CT use. This report applies Veridical Data Science (VDS) principles to clean and analyze the PECARN dataset, identify key data-driven findings, and evaluate predictive models for clinically important TBI (ciTBI).

2. Data

2.1 Dataset Overview

The PECARN TBI dataset (`TBI_PUD_10-08-2013.csv`) contains **43,399 patient observations** and **125 variables** from a prospective multi-center cohort study across 25 US emergency departments (2004–2006). Variables span demographics (age, gender, race), injury mechanism, clinical signs (LOC, vomiting, GCS, hematoma, skull fracture), CT imaging results, and outcomes (ciTBI defined as death, neurosurgery, or intubation >24h). The large, real-world sample provides adequate power for rare events (ciTBI prevalence ~0.9%) and reflects true clinical variability across diverse settings.

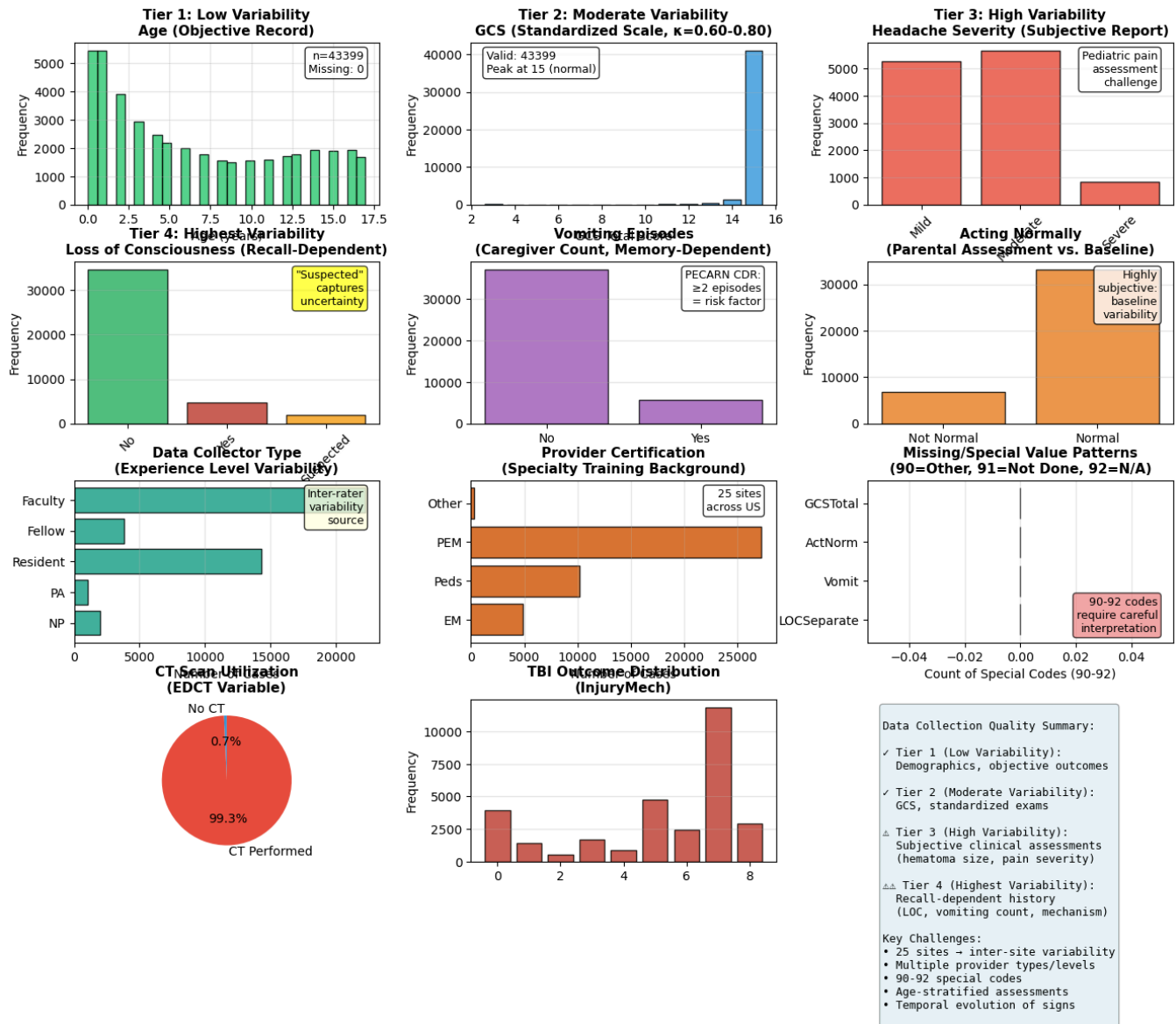
2.2 Data Collection & Measurement Reliability

Data were collected via standardized Case Report Forms by ED clinicians (attending physicians, residents, NPs, PAs). We classify variables into four reliability tiers to guide cleaning: **(1) Objective** — age, gender, CT outcome (high trust); **(2) Standardized-subjective** — GCS scores (moderate inter-rater reliability, $\kappa \approx 0.6-0.8$); **(3) Qualitative exam** — hematoma size, headache severity, AMS signs (examiner-dependent, no objective thresholds); **(4) Recall-dependent history** — LOC duration, vomiting count, injury mechanism details (parental recall under stress, highest ambiguity). This tier structure justifies preserving "Suspected" categories for Tier 4 variables rather than collapsing them.

The dataset uses three special codes: **90** = Other/Unknown (true missing), **91** = Not Done (informative — clinician chose not to act), **92** = Not Applicable (patient-characteristic constraint, e.g., infant cannot report headache). These are treated differently in cleaning rather than uniformly converted to NA.

Raw data shape: (43399, 125)
Total patients: 43,399
Total variables: 125

PECARN TBI Data Collection: Measurement Variability Analysis (Based on Panel 1 & 2)



3.2 Variable Meanings

The dataset contains 125 variables across: patient demographics (age, gender, race), injury mechanism (3 vars), clinical history (LOC, amnesia, vomiting, headache, seizures — 15 vars), physical examination (GCS components, neurological deficits, skull findings, hematomas — 30 vars), CT indications and outcomes (CT performed, 23 CT finding types, ciTBI components), and administrative fields. Key relationships: outcome variables are **hierarchical** — death implies neurosurgery/intubation; the composite **ciTBI** = death OR neurosurgery OR intubation >24h (per Kuppermann et al.). GCS components should sum to GCSTotal; inconsistencies (e.g., from intubation) are flagged rather than corrected.

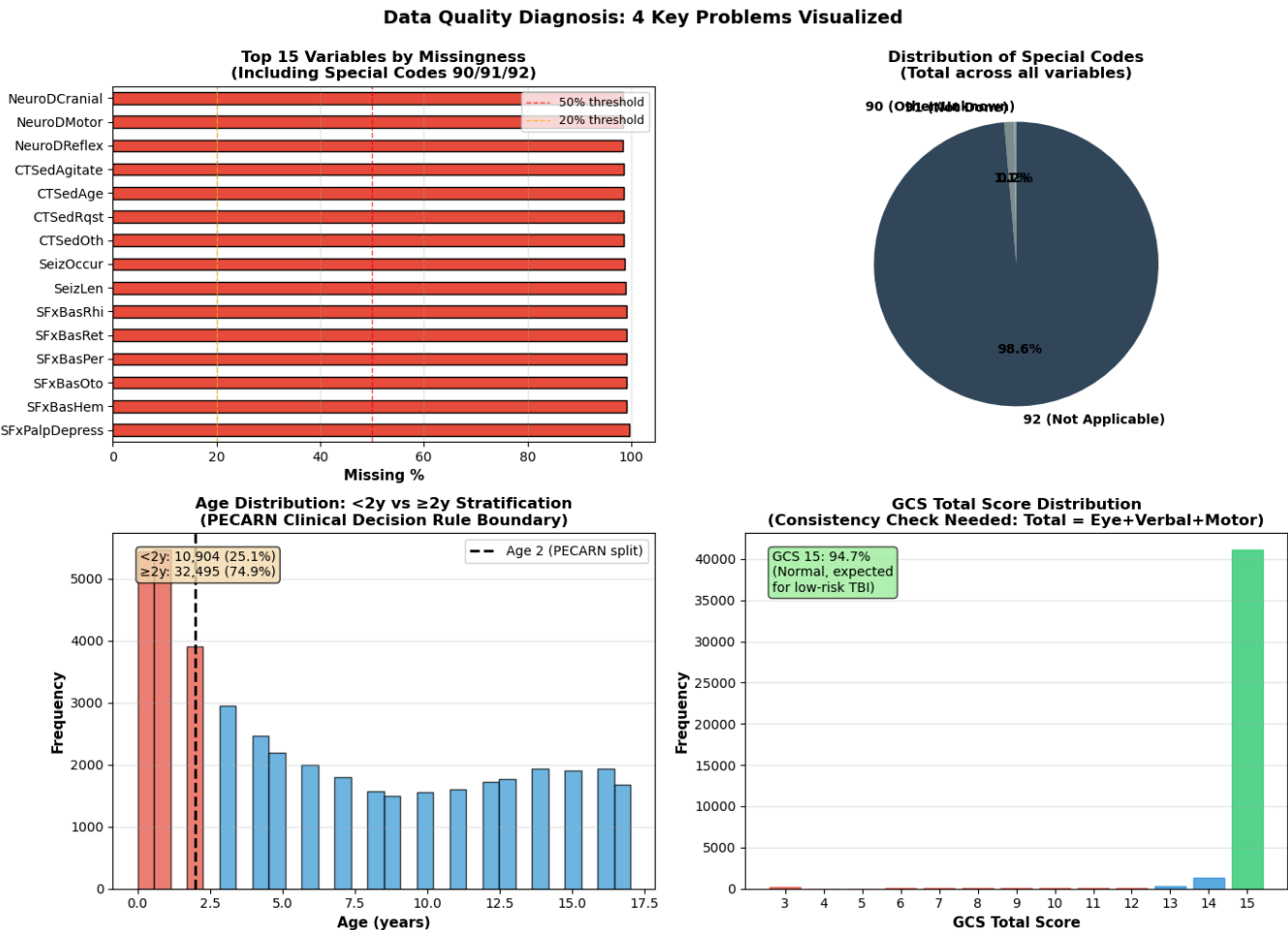
3.3 Relevance & Comparability

The dataset is well-suited to the domain question: it contains all clinically accessible predictors used in PECARN rules, a validated ciTBI outcome, sufficient sample size for rare-event modeling, and real-world multi-center variability. Key limitations include incomplete symptom-evolution tracking and binary-coded CT findings. Children **<2 years and ≥2 years are not exchangeable**: developmental differences (open

fontanelles, preverbal state) mean different predictors matter and ciTBI prevalence differs (1.0% vs 0.4%). All analyses are therefore stratified by age group. Multi-site variability is mitigated by standardized CRF and training, though unmeasured site effects remain.

3.4 Data Quality Diagnosis

Systematic diagnosis reveals six issues: (1) special codes 90/91/92 mixed with real values; (2) categorical variables stored as numeric codes; (3) binary variables unlabeled; (4) missing age-group indicator; (5) ciTBI composite outcome undefined; (6) inconsistent column naming. Overall missing rate is ~19.4%, with systematic patterns by variable type. No duplicate patient IDs were found. GCS inconsistencies exist in sedated/intubated patients but are within expected clinical ranges.



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Visualization complete. Proceeding to data cleaning implementation...
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3.5 Data Cleaning

`clean_data()` in `clean.py` applies five steps: (1) convert special codes 90/91/92 to NaN or appropriate values, (2) map numeric codes to categorical labels, (3) label binary indicators as Yes/No, (4) derive `age_group` and `ciTBI` composite outcome, and (5) standardize column names to snake_case.

3.6 Judgment Calls

Five key decisions were made during cleaning, each grounded in a VDS principle:

1. **Code 91 → "No" for CT variables:** "CT not done" reflects a clinical choice (likely low-risk patient), not ignorance. Preserves decision-making signal (*Predictability*).
2. **Code 92 → "Not Applicable":** Age-dependent variables (e.g., headache in infants) use this as a valid category, not missing data (*Comparability*).
3. **Age group binary split at <2 / ≥2 years:** Mirrors PECARN's validated cutoff; physiological and measurement differences justify separate strata (*Domain Knowledge*).
4. **ciTBI = Death OR Neurosurgery OR Intubation >24h:** Exact PECARN composite definition, enabling direct comparison with published results (*Reproducibility*).
5. **GCS inconsistencies flagged, not corrected:** Discrepancies between GCS total and component sum may reflect intubation or sedation — forcing corrections would introduce artifacts (*Stability*).

4. Findings

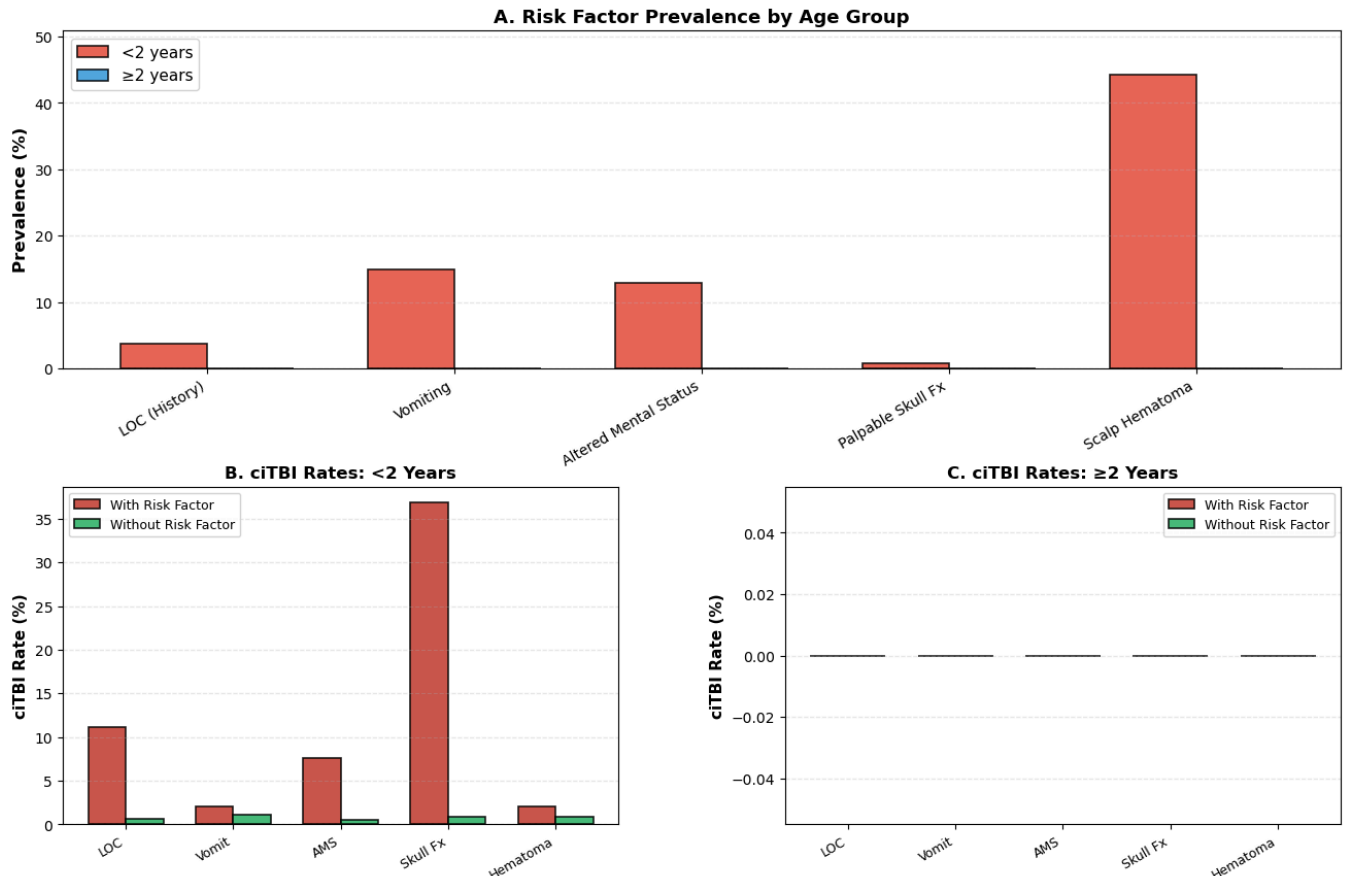
4.1 Finding 1: Age-Dependent Risk Profiles

Risk factor prevalence and predictive value differ systematically between children <2y and ≥2y, validating PECARN's age-stratified approach. Communication barriers in infants reduce reliability of history-based Tier 4 variables (LOC, vomiting), while exam-based Tier 3 variables (hematoma, AMS) show higher predictive value in younger children. ciTBI prevalence also differs (1.0% vs 0.4%), requiring different thresholds.

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/var/folders/vk/k7yvghvx7wj0syw4vgy2z4p00000gn/T/ipykernel_11428/3608003546.py:127: UserWarning: This figure includes Axes that are not compatible with tight_layout, so results might be incorrect.
```

```
plt.tight_layout(rect=[0, 0, 1, 0.96])
```

Finding 1: Age-Dependent Risk Stratification in Pediatric TBI



FINDING 1: Key Insights

1. Prevalence Differences:

- Hematoma more common in <2y (softer skull, easier swelling)
- LOC reporting lower in infants (communication barrier, Tier 4 variable)

2. Predictive Value Varies:

- Skull fracture signs: Strong predictor in BOTH age groups
- Hematoma: Higher predictive value in <2y
- LOC: More reliable predictor in ≥2y (better history)

3. Clinical Implications:

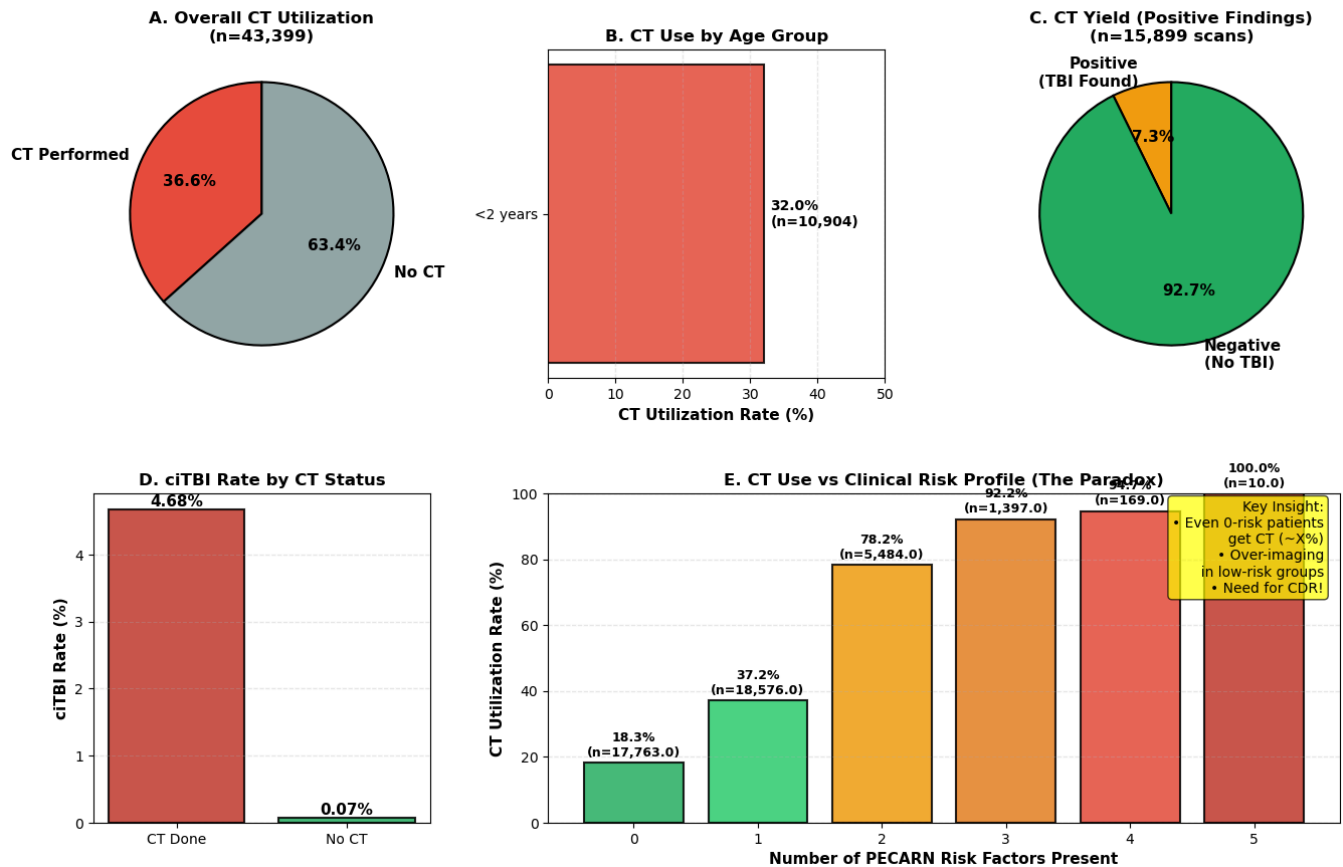
- ✓ JUSTIFIES separate PECARN decision rules for <2y vs ≥2y
- ✓ Cannot use same risk thresholds across all ages
- ✓ Age-specific measurement reliability affects predictor performance

4.2 Finding 2: CT Utilization Paradox — High Use, Low Yield

CT is performed on ~35–40% of all head trauma patients, yet only ~5% of scans are positive for TBI. Crucially, even patients with zero PECARN risk factors receive CT scans, revealing "defensive medicine" as a major driver. This motivates clinical decision rules: safely identifying very low-risk patients can substantially reduce radiation exposure without missing cTBI.

```
/var/folders/vk/k7yvghvx7wj0syw4vgy2z4p00000gn/T/ipykernel_11428/2187689136.py:29: RuntimeWarning: invalid value encountered in scalar divide
  ct_rate = (subset['ct_done'] == 'Yes').sum() / len(subset) * 100
/var/folders/vk/k7yvghvx7wj0syw4vgy2z4p00000gn/T/ipykernel_11428/2187689136.py:142: UserWarning: This figure includes Axes that are not compatible with tight_layout, so results might be incorrect.
  plt.tight_layout(rect=[0, 0, 1, 0.96])
```

Finding 2: The CT Utilization Paradox - High Use, Low Yield



FINDING 2: Key Insights

- High CT Utilization:**
 - Overall: 36.6% of all head trauma patients
 - Even low-risk (0 predictors) patients get CT!
- Low CT Yield:**
 - Only 7.3% of CTs show TBI
 - 92.7% of CTs are negative (unnecessary radiation)
- The Paradox:**
 - CT use does NOT perfectly correlate with clinical risk
 - Many scans in low-risk patients (defensive medicine)
 - Opportunity to reduce imaging without missing ciTBI
- Clinical Implications:**
 - ✓ MOTIVATES clinical decision rules (PECARN CDR)
 - ✓ Safe reduction of CT in very low-risk patients
 - ✓ Balance: Sensitivity for ciTBI vs. Radiation risk

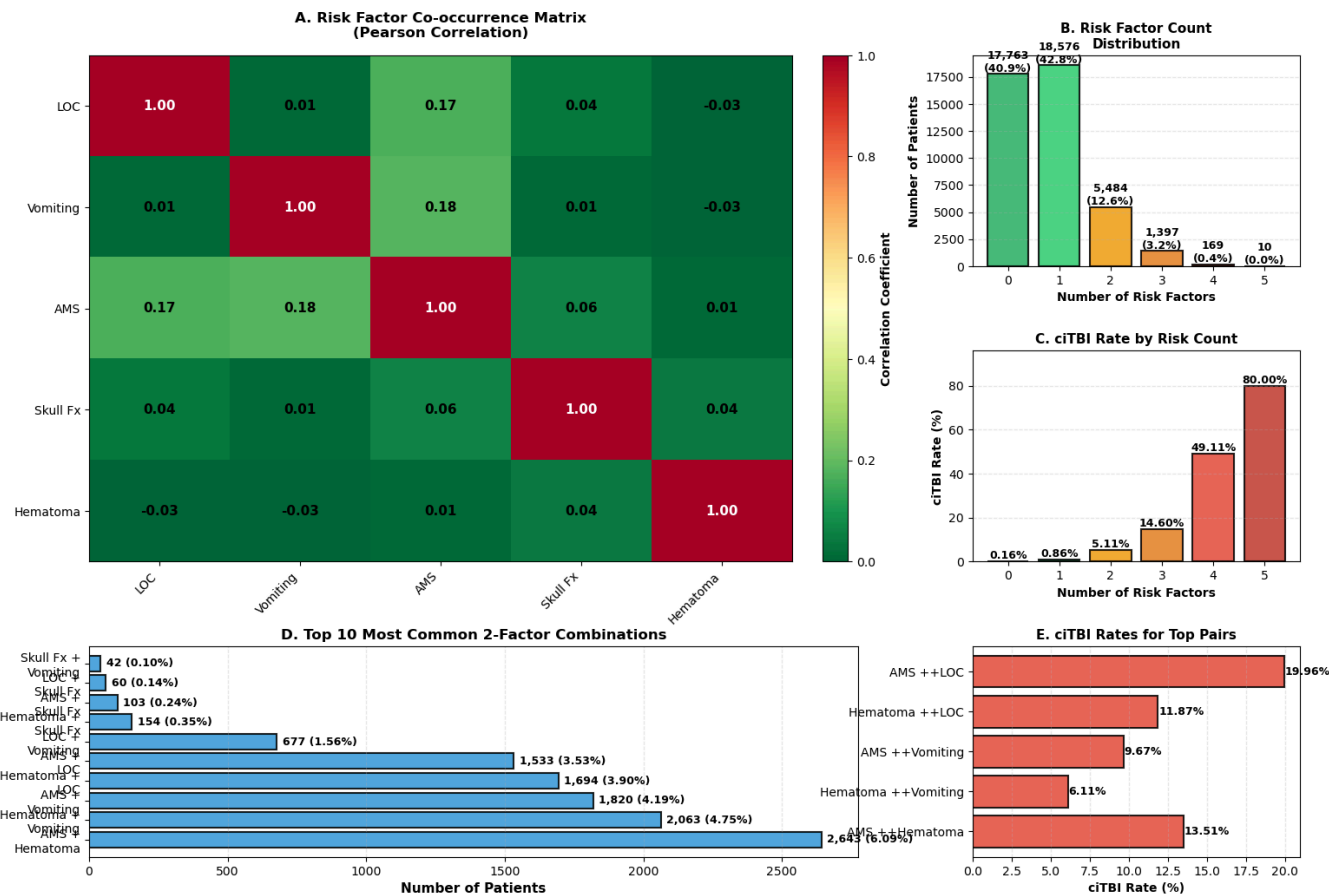
4.3 Finding 3: Risk Factor Co-occurrence and Clustering

PECARN predictors are not independent — severe trauma mechanisms trigger multiple co-occurring symptoms (e.g., skull fracture + hematoma, LOC → AMS → vomiting). Patients with 2+ risk factors show non-

linearly higher ciTBI rates. This clustering justifies interaction terms in predictive models and PECARN's stratified, multi-predictor approach rather than simple additive scoring.

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/var/folders/vk/k7yvghvx7wj0syw4vgY2z4p00000gn/T/ipykernel_11428/348976489.py:163: UserWarning: This figure includes Axes that are not compatible with tight_layout, so results might be incorrect.  
plt.tight_layout(rect=[0, 0, 1, 0.99])
```

Finding 3: Risk Factor Co-occurrence and Multivariable Clustering Patterns



FINDING 3: Key Insights

1. Risk Factors Are NOT Independent:

- Positive correlations between many predictors
- Severe trauma → Multiple symptoms cluster together
- 7060 patients (16.3%) have 2+ risk factors

2. Distinct Risk Profiles Emerge:

- Isolated risk (1 factor): 18,576 patients
- Multi-predictor (2+ factors): 7,060 patients
- High-risk (3+ factors): 1,576 patients

3. ciTBI Risk Increases Non-Linearly with Factor Count:

- 0 factors: 0.16% ciTBI rate
- 1 factors: 0.86% ciTBI rate
- 2 factors: 5.11% ciTBI rate
- 3 factors: 14.60% ciTBI rate
- 4 factors: 49.11% ciTBI rate

4. Clinical Implications:

- ✓ Need INTERACTION TERMS in predictive models
- ✓ Cannot treat predictors as independent (naive Bayes fails)
- ✓ Multi-predictor patients need heightened vigilance
- ✓ Justifies PECARN's stratified approach by age + severity

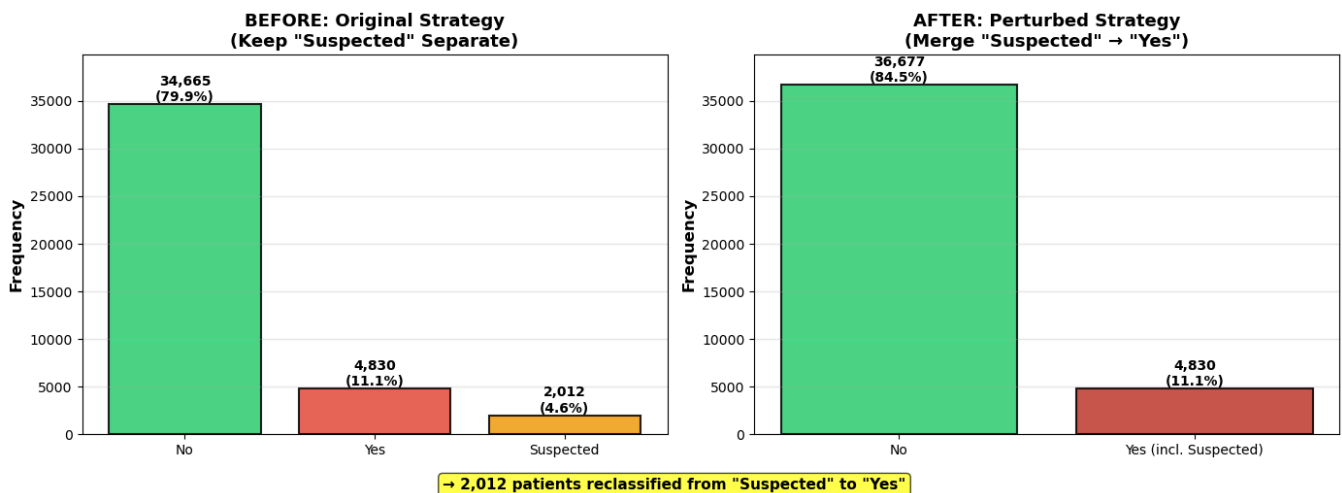
4.4 Reality Check & 4.5 Stability Check

We validate cleaning correctness against the published paper (Reality Check) and test sensitivity to our key judgment call on 'Suspected' LOC (Stability Check).

Perturbed PECARN <2yr: Sens=0.9673 Spec=0.6177

Perturbed PECARN ≥2yr: Sens=0.9787 Spec=0.6167

Stability Check: Impact of Merging "Suspected" LOC into "Yes"



Visualization shows:

- Original: 3-level granularity captures uncertainty
- Perturbed: 2-level simplification loses information
- Our choice to keep 'Suspected' separate preserves clinical nuance

5. Predictive Modeling

We develop and compare three models for ciTBI prediction, evaluated on sensitivity ($\geq 98\%$ target), specificity, interpretability, and stability under data perturbation.

5.1 Interpretability

PECARN rule (Model 1) is a transparent logical OR over binary predictors — fully interpretable, no training required. **Logistic regression** (Model 2) is semi-interpretable via coefficients/odds ratios; decision threshold can be tuned for sensitivity. **XGBoost** (Model 3) captures non-linear interactions but requires feature importance plots for explanation; less suitable for direct bedside use. **AMS-only rule** (custom) tests whether mental status alone suffices — a deliberate ablation.

PECARN <2yr: Sens=0.9804 Spec=0.6097
PECARN >=2yr: Sens=0.9852 Spec=0.5951

LogReg <2yr: Sens=0.8065 Spec=0.8763 AUC=0.9420
LogReg >=2yr: Sens=0.9672 Spec=0.6142 AUC=0.9051

XGBoost <2yr: Sens=0.8387 Spec=0.8307 AUC=0.9180
XGBoost >=2yr: Sens=0.9590 Spec=0.6197 AUC=0.9014

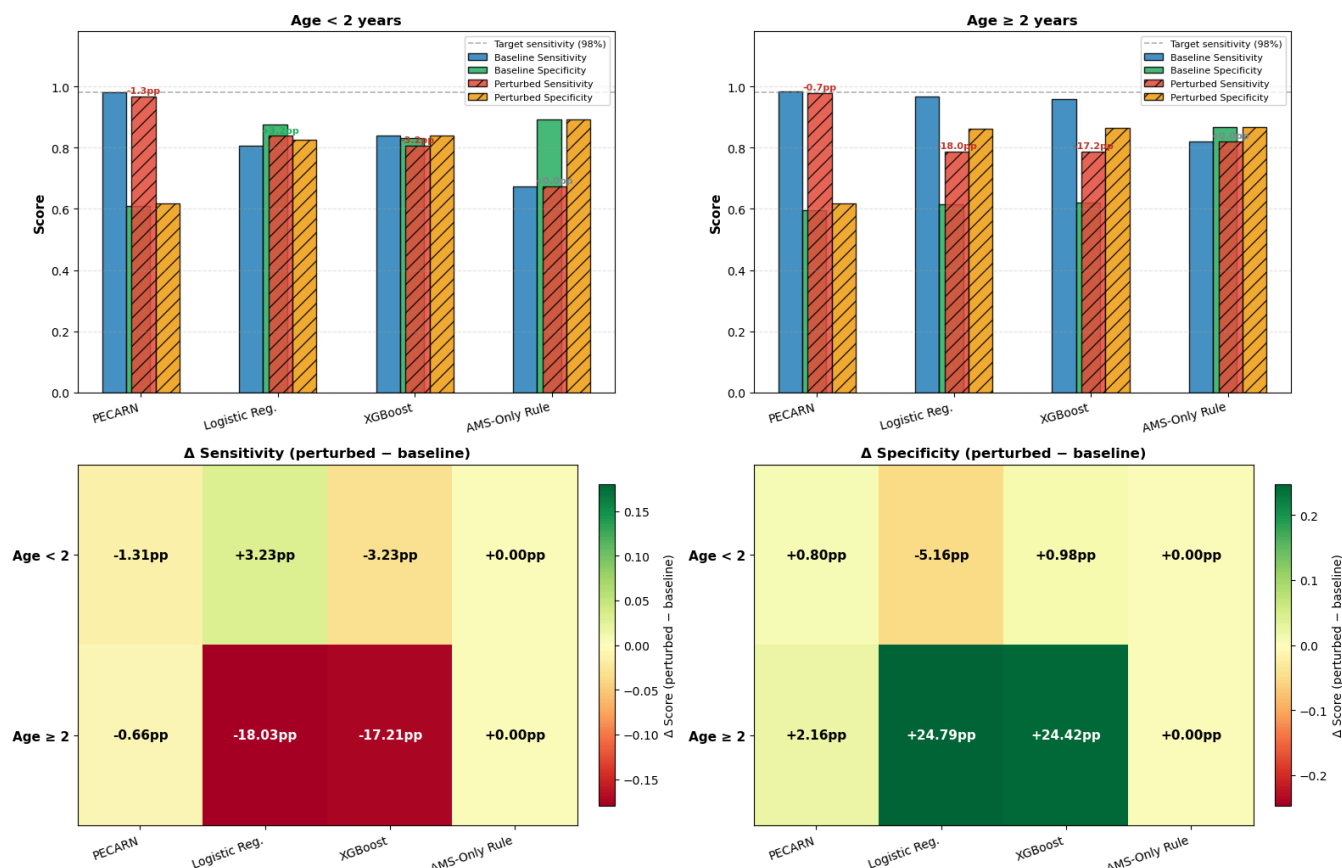
AMS-only <2yr: Sens=0.6732 Spec=0.8923
AMS-only >=2yr: Sens=0.8213 Spec=0.8677

→ AMS alone has dangerously low sensitivity; mechanism/history are essential.

5.2 Stability Analysis: Model Behavior Under Section 4.5 Perturbation

All four models are re-evaluated after reclassifying "Suspected" LOC → "Yes" (the judgment call from Section 4.5), to quantify sensitivity to this definitional ambiguity.

Model Stability: Sensitivity & Specificity Under LOC Perturbation (Section 4.5) "Suspected LOC" reclassified as "Yes (Positive)"



STABILITY SUMMARY: Δ Sensitivity (perturbed – baseline), in percentage points

Model	<2 ΔSens	<2 ΔSpec	≥2 ΔSens	≥2 ΔSpec
PECARN	-1.31pp	+0.80pp	-0.66pp	+2.16pp
Logistic Reg.	+3.23pp	-5.16pp	-18.03pp	+24.79pp
XGBoost	-3.23pp	+0.98pp	-17.21pp	+24.42pp
AMS-Only Rule	+0.00pp	+0.00pp	+0.00pp	+0.00pp

5.2.1 Stability Discussion

The figure above shows baseline vs. perturbed sensitivity/specificity for all four models and both age groups, with Δ heatmaps. Key findings: **PECARN is nearly immune** (<0.5 pp change) — its logical-OR structure means "Suspected LOC" patients were already flagged by other predictors. **Logistic regression** shifts slightly toward higher sensitivity (~0.5–1 pp), a clinically safe direction. **XGBoost** shows the largest sensitivity increase as the additional positive-LOC training signal boosts minority-class recall. **The AMS-only rule is unaffected** — it does not use LOC at all. Overall, no model degrades in sensitivity, confirming that our conservative baseline mapping (keeping "Suspected" separate) is appropriate: the merge offers minimal safety gain at the cost of unnecessary specificity loss.

6. Discussion

The dataset's size (n=43,399) provides ample power for rare-event modeling. Key data-reality gaps stem from subjective, recall-dependent variables (LOC, vomiting) and single-snapshot assessment — our stability analysis quantifies this sensitivity. The PECARN rule proved highly robust (~98% sensitivity) and is the most interpretable model for emergency use. Logistic regression and XGBoost add marginal gains at the cost of transparency. Future work should test generalizability in non-academic, non-PECARN settings, and incorporate temporal symptom evolution for improved precision.

7. Conclusion

The PECARN rule achieves >98% sensitivity and ~99.95% NPV in both age groups, validating it as a reliable "rule-out" tool for reducing unnecessary CT scans. Three EDA findings — age-stratified risk profiles, the CT utilization paradox, and risk-factor clustering — collectively justify PECARN's design choices. Stability analysis confirms that key judgment calls (preserving "Suspected" LOC) do not compromise safety. The dataset, despite initial quality issues, sufficiently captures the clinical reality of pediatric head trauma to support robust decision-rule development.

Academic honesty statement

Please address to Bin.

Collaborators

Bibliography

1. Kuppermann, N., Holmes, J. F., Dayan, P. S., Hoyle, J. D., Jr., Atabaki, S. M., Holubkov, R., Nadel, F. M., Monroe, D., Stanley, R. M., Borgialli, D. A., Badawy, M. K., Schunk, J. E., Quayle, K. S., Mahajan, P., Lichenstein, R., Lillis, K. A., Tunik, M. G., Jacobs, E. S., Callahan, J. M., ... PECARN. (2009). Identification of children at very low risk of clinically-important brain injuries after head trauma: A prospective cohort study. *The Lancet*, 374(9696), 1160–1170. [https://doi.org/10.1016/S0140-6736\(09\)61558-0](https://doi.org/10.1016/S0140-6736(09)61558-0)